Potential Solutions & Risk Mitigation

GUIDELINE SLIDE

Potential Solutions (design choices that prevent the problem)

Thrust vs basket drag:
baffle gaps sized to the largest debris, motors on swappable mounts

Baffle jamming:
slot-in, adjustable baffle stages (not glued)

Instability:
central battery, centred basket, floats with reserve buoyancy

Battery runtime:
3S pack sized to estimated draw

Camera/control link:
separate regulated 5 V supply for the ESP32-CAM, Arduino drives motors independently

Propeller entanglement:
propeller guard

Risk Mitigation (backup technical solutions)

Low thrust:
add a gear-reduction stage or higher-torque motor, or revert to a plain V-intake with a fixed foil lip

Baffle jams or adds drag:
adjust fin angle and gap, or drop to the foil-lip-only design

Uneven loading:
reposition ballast; floats are never shrunk to save cost

Short runtime:
add a second parallel cell (+₹300–400)

Link drops:
drive by line of sight at short range, recover with the tether

Propeller tangles:
switch to paddle-wheel propulsion